```
#José Alejandro Cabeza Parra
# Introduccion

#Este proyecto analiza el comportamiento de ventas de una empresa de distribución musical con presencia
#internacional, con el objetivo de identificar los mercados, artistas, clientes y géneros que generan mayor rentabilidad.

# Objetivo

#Identificar:

#Países con mayor generación de ingresos
#Artistas y géneros más rentables
#Clientes con mayor volumen de compra
```

```
import sqlite3
import pandas as pd
```

```
from google.colab import files
uploaded = files.upload()
```

Elegir archivos Chinook.db

Chinook.db(n/a) - 1067008 bytes, last modified: 10/11/2025 - 100% done

```
Saving Chinook.db to Chinook (1).db
```

```
conn= sqlite3.connect('Chinook.db')
```

```
query = "SELECT name FROM sqlite_master WHERE type='table';"
df_tablas =pd.read_sql_query(query,conn)
```

```
df_tablas
```

	name
0	Album
1	Artist
2	Customer
3	Employee
4	Genre
5	Invoice
6	InvoiceLine
7	MediaType
8	Playlist
9	PlaylistTrack
10	Track

Pasos siguientes: Generar código con df_tablas New interactive sheet

```
pd_customer= pd.read_sql_query("PRAGMA TABLE_INFO(Customer)",conn)
```

```
pd_album= pd.read_sql_query("PRAGMA TABLE_INFO(Album)",conn)
```

```
pd_artist=pd.read_sql_query("PRAGMA TABLE_INFO(Artist)",conn)
```

```
pd_employee=pd.read_sql_query("PRAGMA TABLE_INFO(Employee)",conn)
```

```
pd_genre= pd.read_sql_query("PRAGMA TABLE_INFO(Genre)",conn)
```

```
pd_invoice= pd.read_sql_query("PRAGMA TABLE_INFO(Invoice)",conn)
```

```
pd_invoiceline= pd.read_sql_query("PRAGMA TABLE_INFO(InvoiceLine)",conn)
```

```
pd_mediatype= pd.read_sql_query("PRAGMA TABLE_INFO(MediaType)",conn)
```

```
pd_playlist= pd.read_sql_query("PRAGMA TABLE_INFO(Playlist)",conn)
```

```
pd_playlisttrack= pd.read_sql_query("PRAGMA TABLE_INFO(PlaylistTrack)",conn)
```

```
pd_track= pd.read_sql_query("PRAGMA TABLE_INFO(Track)",conn)
```

```
dfn_customer = pd.read_sql_query("SELECT * FROM customer LIMIT 5;", conn)
```

```
dfn_customer
```

	CustomerId	FirstName	LastName	Company	Address	City	State	Country	PostalCode	Phone	Fax	
0	1	Luís	Gonçalves	Embraer - Empresa Brasileira de Aeronáutica S.A.	Av. Brigadeiro Faria Lima, 2170	São José dos Campos	SP	Brazil	12227-000	+55 (12) 3923-5555	+55 (12) 3923-5566	luisg@embrae
1	2	Leonie	Köhler	None	Theodor-Heuss-Straße 34	Stuttgart	None	Germany	70174	+49 0711 2842222	None	leonekohler@s
2	3	François	Tremblay	None	1498 rue Bélanger	Montréal	QC	Canada	H2G 1A7	+1 (514) 721-4711	None	ftremblay@gr
3	4	Bjørn	Hansen	None	Ullevålsveien	Oslo	None	Norway	0171	+47 22	None	bjorn.hansen@u

Pasos siguientes: Generar código con `dfn_customer` New interactive sheet

```
dfn_album = pd.read_sql_query("SELECT * FROM Album LIMIT 5;", conn)
```

```
dfn_artist = pd.read_sql_query("SELECT * FROM Artist LIMIT 5;", conn)
```

```
dfn_employee = pd.read_sql_query("SELECT * FROM Employee LIMIT 5;", conn)
```

```
dfn_genre = pd.read_sql_query("SELECT * FROM Genre LIMIT 5;", conn)
```

```
dfn_invoice = pd.read_sql_query("SELECT * FROM Invoice LIMIT 5;", conn)
```

```
dfn_invoiceline = pd.read_sql_query("SELECT * FROM InvoiceLine LIMIT 5;", conn)
```

```
dfn_mediatype = pd.read_sql_query("SELECT * FROM MediaType LIMIT 5;", conn)
```

```
dfn_playlist = pd.read_sql_query("SELECT * FROM Playlist LIMIT 5;", conn)
```

```
dfn_playlisttrack = pd.read_sql_query("SELECT * FROM PlaylistTrack LIMIT 5;", conn)
```

```
dfn_track = pd.read_sql_query("SELECT * FROM track LIMIT 5;", conn)
```

```
pd.read_sql_query("PRAGMA Foreign_Key_list(Customer)",conn)#en la tabla custumer la columna supportrepid
#hace referencia a la columna employeedid de la tabla employeed
```

	id	seq	table	from	to	on_update	on_delete	match
0	0	0	Employee	SupportRepId	EmployeeId	NO ACTION	NO ACTION	NONE

```
dfn_artist
```

	ArtistId	Name
0	1	AC/DC
1	2	Accept
2	3	Aerosmith
3	4	Alanis Morissette
4	5	Alice In Chains

Pasos siguientes: Generar código con `dfn_artist` New interactive sheet

```
pd.read_sql_query("PRAGMA Foreign_Key_list(Artist)",conn)
```

id	seq	table	from	to	on_update	on_delete	match

```
pd.read_sql_query("PRAGMA Foreign_Key_list(Album)",conn)
```

	id	seq	table	from	to	on_update	on_delete	match
0	0	0	Artist	ArtistId	ArtistId	NO ACTION	NO ACTION	NONE

```
dfn_album
```

	AlbumId	Title	ArtistId
0	1	For Those About To Rock We Salute You	1
1	2	Balls to the Wall	2
2	3	Restless and Wild	2
3	4	Let There Be Rock	1
4	5	Big Ones	3

Pasos siguientes: Generar código con `dfn_album` New interactive sheet

```
dfn_customer
```

	CustomerId	FirstName	LastName	Company	Address	City	State	Country	PostalCode	Phone	Fax	
0	1	Luís	Gonçalves	Embraer - Empresa Brasileira de Aeronáutica S.A.	Av. Brigadeiro Faria Lima, 2170	São José dos Campos	SP	Brazil	12227-000	+55 (12) 3923-5555	+55 (12) 3923-5566	luisg@embrae
1	2	Leonie	Köhler	None	Theodor-Heuss-Straße 34	Stuttgart	None	Germany	70174	+49 0711 2842222	None	leonekohler@s
2	3	François	Tremblay	None	1498 rue Bélanger	Montréal	QC	Canada	H2G 1A7	+1 (514) 721-4711	None	ftremblay@gr
3	4	Bjørn	Hansen	None	Ullevålsveien	Oslo	None	Norway	0171	+47 22	None	bjorn.hansen@u

Pasos siguientes: Generar código con `dfn_customer` New interactive sheet

```
dfn_invoice
```

	InvoiceId	CustomerId	InvoiceDate	BillingAddress	BillingCity	BillingState	BillingCountry	BillingPostalCode	Total
0	1	2	2009-01-01 00:00:00	Theodor-Heuss-Straße 34	Stuttgart	None	Germany	70174	1.98
1	2	4	2009-01-02 00:00:00	Ullevålsveien 14	Oslo	None	Norway	0171	3.96
2	3	8	2009-01-03 00:00:00	Grétrystraat 63	Brussels	None	Belgium	1000	5.94

Pasos siguientes: Generar código con dfn_invoice New interactive sheet

```
dfn_invoiceline
```

	InvoiceLineId	InvoiceId	TrackId	UnitPrice	Quantity
0	1	1	2	0.99	1
1	2	1	4	0.99	1
2	3	2	6	0.99	1
3	4	2	8	0.99	1
4	5	2	10	0.99	1

Pasos siguientes: Generar código con dfn_invoiceline New interactive sheet

```
dfn_track
```

	TrackId	Name	AlbumId	MediaTypeId	GenreId	Composer	Milliseconds	Bytes	UnitPrice
0	1	For Those About To Rock (We Salute You)	1	1	1	Angus Young, Malcolm Young, Brian Johnson	343719	11170334	0.99
1	2	Balls to the Wall	2	2	1	None	342562	5510424	0.99
2	3	Fast As a Shark	3	2	1	F. Baltes, S. Kaufman, U. Dirkscneider & W. Ho...	230619	3990994	0.99
3	4	Restless and Wild	3	2	1	F. Baltes, R.A. Smith-Diesel,	252051	4331779	0.99

Pasos siguientes: Generar código con dfn_track New interactive sheet

```
dfn_genre
```

	GenreId	Name
0	1	Rock
1	2	Jazz
2	3	Metal
3	4	Alternative & Punk
4	5	Rock And Roll

Pasos siguientes: Generar código con dfn_genre New interactive sheet

```
#identificar cuales generos y artistas generan mas ingresos y quienes son los mejores clientes

relaciones= []
for t in df_tablas["name"]:
  fk= pd.read_sql_query(f"PRAGMA Foreign_Key_list({t})",conn)
  if not fk.empty:
    fk["tabla_origen"]=t
    relaciones.append(fk)
```

```
print(relaciones)
```

```
[   id  seq   table      from        to  on_update  on_delete match  \
0   0    0  Artist  ArtistId  ArtistId  NO ACTION  NO ACTION  NONE

  tabla_origen
0        Album  ,    id  seq     table          from          to  on_update  on_delete match  \
0   0    0  Employee  SupportRepId  EmployeeId  NO ACTION  NO ACTION  NONE

  tabla_origen
0     Customer  ,    id  seq     table       from          to  on_update  on_delete match  \
0   0    0  Employee  ReportsTo  EmployeeId  NO ACTION  NO ACTION  NONE

  tabla_origen
0     Employee  ,    id  seq     table        from          to  on_update  on_delete match  \
0   0    0  Customer  CustomerId  CustomerId  NO ACTION  NO ACTION  NONE

  tabla_origen
0      Invoice  ,    id  seq    table       from         to  on_update  on_delete match  \
0   0    0    Track    TrackId    TrackId  NO ACTION  NO ACTION  NONE
1   1    0  Invoice  InvoiceId  InvoiceId  NO ACTION  NO ACTION  NONE

  tabla_origen
0  InvoiceLine
1  InvoiceLine  ,    id  seq     table        from          to  on_update  on_delete match  \
0   0    0     Track     TrackId     TrackId  NO ACTION  NO ACTION  NONE
1   1    0  Playlist  PlaylistId  PlaylistId  NO ACTION  NO ACTION  NONE

    tabla_origen
0  PlaylistTrack
1  PlaylistTrack  ,    id  seq      table         from           to  on_update  on_delete match  \
0   0    0  MediaType  MediaTypeId  MediaTypeId  NO ACTION  NO ACTION  NONE
1   1    0      Genre      GenreId      GenreId  NO ACTION  NO ACTION  NONE
2   2    0      Album      AlbumId      AlbumId  NO ACTION  NO ACTION  NONE

  tabla_origen
0        Track
1        Track
2        Track  ]
```

```
df_rel= pd.concat(relaciones,ignore_index=True) if relaciones else pd.DataFrame()
```

```
if not df_rel.empty:
  df_rel = df_rel.rename(columns={
      "table":"tabla_destino",
      "from":"columna_origen",
      "to":"columna_destino"
      })[["tabla_origen","tabla_destino","columna_origen","columna_destino"]]
```

```
print(df_rel)
```

```
     tabla_origen tabla_destino columna_origen columna_destino
0           Album        Artist       ArtistId        ArtistId
1        Customer      Employee   SupportRepId      EmployeeId
2        Employee      Employee      ReportsTo      EmployeeId
3         Invoice      Customer     CustomerId      CustomerId
4     InvoiceLine         Track        TrackId         TrackId
5     InvoiceLine       Invoice      InvoiceId       InvoiceId
6   PlaylistTrack         Track        TrackId         TrackId
7   PlaylistTrack      Playlist     PlaylistId      PlaylistId
8           Track     MediaType    MediaTypeId     MediaTypeId
9           Track         Genre        GenreId         GenreId
10          Track         Album        AlbumId         AlbumId
```

```
q1_generos="""SELECT g.Name AS Genero, SUM(il.UnitPrice * il.Quantity) AS Ingresos
FROM InvoiceLine il
JOIN Track t ON il.TrackId = t.TrackId
JOIN Genre g ON t.GenreId = g.GenreId
GROUP BY Genero
ORDER BY Ingresos DESC
"""
df_generos_ingresos= pd.read_sql_query(q1_generos,conn)
df_generos_ingresos
```

	Genero	Ingresos
0	Rock	826.65
1	Latin	382.14
2	Metal	261.36
3	Alternative & Punk	241.56
4	TV Shows	93.53
5	Jazz	79.20
6	Blues	60.39
7	Drama	57.71
8	R&B/Soul	40.59
9	Classical	40.59
10	Sci Fi & Fantasy	39.80
11	Reggae	29.70
12	Pop	27.72
13	Soundtrack	19.80
14	Comedy	17.91
15	Hip Hop/Rap	16.83
16	Bossa Nova	14.85
17	Alternative	13.86
18	World	12.87
19	Science Fiction	11.94
20	Heavy Metal	11.88
21	Electronica/Dance	11.88
22	Easy Listening	9.90
23	Rock And Roll	5.94

Pasos siguientes: Generar código con df_generos_ingresos New interactive sheet

```
#identificar cual artistas genera mas ingresos y quienes son los mejores clientes
q1_artista="""SELECT a.Name AS Artista, SUM(il.UnitPrice * il.Quantity) AS Ingresos
FROM InvoiceLine il
JOIN Track t ON il.TrackId = t.TrackId
JOIN Album al ON t.AlbumId = al.AlbumId
JOIN Artist a ON al.ArtistId = a.ArtistId
GROUP BY Artista
ORDER BY Ingresos DESC
LIMIT 10;
"""
df_artista_ingresos= pd.read_sql_query(q1_artista,conn)
df_artista_ingresos
```

	Artista	Ingresos
0	Iron Maiden	138.60
1	U2	105.93
2	Metallica	90.09
3	Led Zeppelin	86.13
4	Lost	81.59
5	The Office	49.75
6	Os Paralamas Do Sucesso	44.55
7	Deep Purple	43.56
8	Faith No More	41.58
9	Eric Clapton	39.60

Pasos siguientes: Generar código con df_artista_ingresos New interactive sheet

```
#mejores clientes

q3_clientes="""SELECT c.FirstName || ' ' || c.LastName AS Cliente, SUM(i.Total) AS Total_Compras
FROM Customer c
JOIN Invoice i ON c.CustomerId = i.CustomerId
GROUP BY Cliente
ORDER BY Total_Compras DESC
LIMIT 10;
"""
df_Cliente_ingresos= pd.read_sql_query(q3_clientes,conn)
df_Cliente_ingresos
```

	Cliente	Total_Compras
0	Helena Holý	49.62
1	Richard Cunningham	47.62
2	Luis Rojas	46.62
3	Ladislav Kovács	45.62
4	Hugh O'Reilly	45.62
5	Julia Barnett	43.62
6	Fynn Zimmermann	43.62
7	Frank Ralston	43.62
8	Victor Stevens	42.62
9	Astrid Gruber	42.62

Pasos siguientes: Generar código con df_Cliente_ingresos New interactive sheet

```
#union de las 3 consultas en una sola

q_master = """SELECT
              i.InvoiceId AS Factura,
              i.InvoiceDate AS Fecha_Factura,
              c.FirstName || ' ' || c.LastName AS Cliente,
              c.Country AS Pais,
              ar.Name AS Artista,
              g.Name AS Genero,
              t.Name AS Cancion,
              il.UnitPrice AS Precio_Unitario,
              il.Quantity AS Cantidad,
              (il.UnitPrice * il.Quantity) AS Total_Linea
          FROM InvoiceLine il
          JOIN Invoice i ON i.InvoiceId=il.InvoiceId
          JOIN Customer c ON c.CustomerId=i.CustomerId
          JOIN Track t ON t.TrackId=il.TrackId
          JOIN Genre g ON g.GenreId=t.GenreId
          JOIN Album al ON al.AlbumId=t.AlbumId
          JOIN Artist ar ON ar.ArtistId=al.ArtistId;
```

```
        """
df_master= pd.read_sql_query(q_master,conn)
df_master.head()
```

	Factura	Fecha_Factura	Cliente	Pais	Artista	Genero	Cancion	Precio_Unitario	Cantidad	Total_Linea
0	1	2009-01-01 00:00:00	Leonie Köhler	Germany	Accept	Rock	Balls to the Wall	0.99	1	0.99
1	1	2009-01-01 00:00:00	Leonie Köhler	Germany	Accept	Rock	Restless and Wild	0.99	1	0.99
2	2	2009-01-02 00:00:00	Bjørn Hansen	Norway	AC/DC	Rock	Put The Finger On You	0.99	1	0.99

Pasos siguientes: Generar código con df_master | New interactive sheet

```
#tipos, nulos, y duplicados
df_master.info()
print("\nNulos por columnas:\n", df_master.isnull().sum())

#Conversion de tipo de dato a fecha
df_master['Fecha_Factura'] = pd.to_datetime(df_master['Fecha_Factura'])

#extraer dia,mes,año para hacer las consultas facilmente
df_master['Dia'] = df_master['Fecha_Factura'].dt.day
df_master['Mes'] = df_master['Fecha_Factura'].dt.month
df_master['Año'] = df_master['Fecha_Factura'].dt.year

#Numero de duplicados exactos si los hubiera
dupli= df_master.duplicated().sum()
print(f"\nDuplicados exactos : {dupli}")

# Eliminamos duplicados si existen
df= df_master.drop_duplicates()
print("Duplicados eliminados. total de filas ahora: ", df.shape[0])

#ahora si alguna fecha no tiene el formato esperado
#vamos a decirle que nos arroje un error NaT
df_master['Fecha_Factura'] = pd.to_datetime(df_master['Fecha_Factura'], errors='coerce')
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Factura          2240 non-null   int64
 1   Fecha_Factura    2240 non-null   object
 2   Cliente          2240 non-null   object
 3   Pais             2240 non-null   object
 4   Artista          2240 non-null   object
 5   Genero           2240 non-null   object
 6   Cancion          2240 non-null   object
 7   Precio_Unitario  2240 non-null   float64
 8   Cantidad         2240 non-null   int64
 9   Total_Linea      2240 non-null   float64
dtypes: float64(2), int64(2), object(6)
memory usage: 175.1+ KB

Nulos por columnas:
 Factura            0
Fecha_Factura      0
Cliente            0
Pais               0
Artista            0
Genero             0
Cancion            0
Precio_Unitario    0
Cantidad           0
Total_Linea        0
dtype: int64

Duplicados exactos : 5
Duplicados eliminados. total de filas ahora:  2235
```

```
# ahora vamos a convertir los tipo string,
#borramos los espacios, y pondremos la primera
```

```
#letra en mayuscula

columnas_string=["Cliente","Pais","Artista","Genero","Cancion"]
for columna in columnas_string:
  df_master[columna]= df_master[columna].astype("string").str.strip().str.title()
```

```
# ahora vamos a convertir los tipo int,

columnas_int=["Factura","Cantidad","Precio_Unitario","Total_Linea"]
for columna in columnas_int:
  df_master[columna]= pd.to_numeric(df_master[columna], errors='coerce' )
```

```
df_master.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 13 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Factura          2240 non-null   int64
 1   Fecha_Factura    2240 non-null   datetime64[ns]
 2   Cliente          2240 non-null   string
 3   Pais             2240 non-null   string
 4   Artista          2240 non-null   string
 5   Genero           2240 non-null   string
 6   Cancion          2240 non-null   string
 7   Precio_Unitario  2240 non-null   float64
 8   Cantidad         2240 non-null   int64
 9   Total_Linea      2240 non-null   float64
 10  Dia              2240 non-null   int32
 11  Mes              2240 non-null   int32
 12  Año              2240 non-null   int32
dtypes: datetime64[ns](1), float64(2), int32(3), int64(2), string(5)
memory usage: 201.4 KB
```

```
# validar que no tenga precios negativos en el dataframe
(df_master["Precio_Unitario"]<0).sum()
```

```
np.int64(0)
```

```
# validar que no hallan cantidades negativas
(df_master["Cantidad"]<0).sum()
```

```
np.int64(0)
```

```
#verificar si el total linea es correcto
df_master["Total_linea_cal"]=df_master['Precio_Unitario']*df_master['Cantidad']
(df_master["Total_Linea"]!=df_master["Total_linea_cal"]).sum()
```

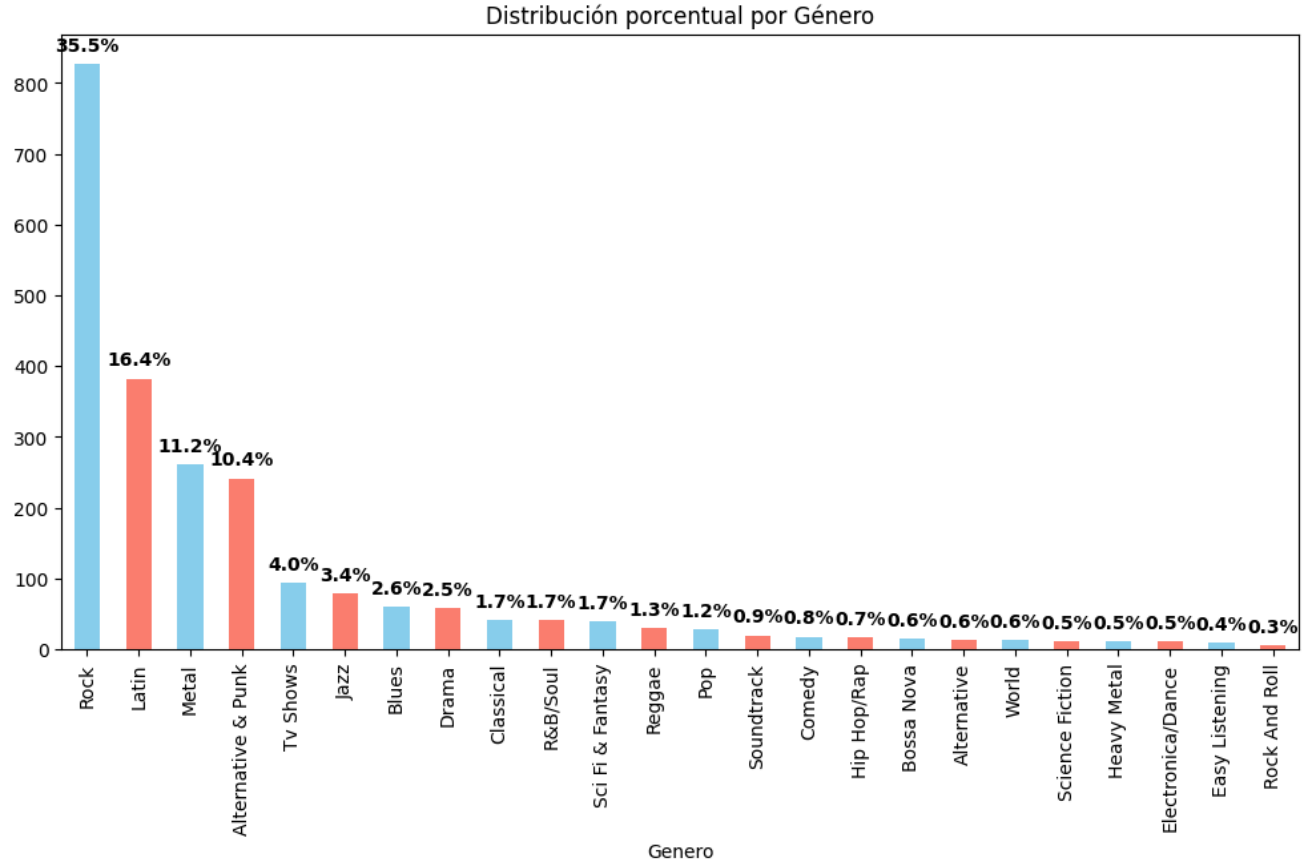
```
np.int64(0)
```

```
columnas_categoria = ["Genero","Artista","Pais"]
for columna in columnas_categoria:
  df_master[columna] = df_master[columna].astype("category")
```

```
#graficamos
import matplotlib.pyplot as plt
# 1. Agrupamos y calculamos el total general
data_genero = df_master.groupby("Genero")["Total_Linea"].sum().sort_values(ascending=False)
total_general = data_genero.sum()

# 2. Creamos el gráfico
ax = data_genero.plot(kind="bar", figsize=(12, 6), color=['skyblue', 'salmon'])

# 3. Iteramos para poner los porcentajes
for p in ax.patches:
    # Calculamos el % (valor de la barra / total)
    porcentaje = f'{(100 * p.get_height() / total_general):.1f}%'

    # Anotamos el texto sobre la barra
    ax.annotate(porcentaje,
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='bottom',
                xytext=(0, 5),
                textcoords='offset points',
```

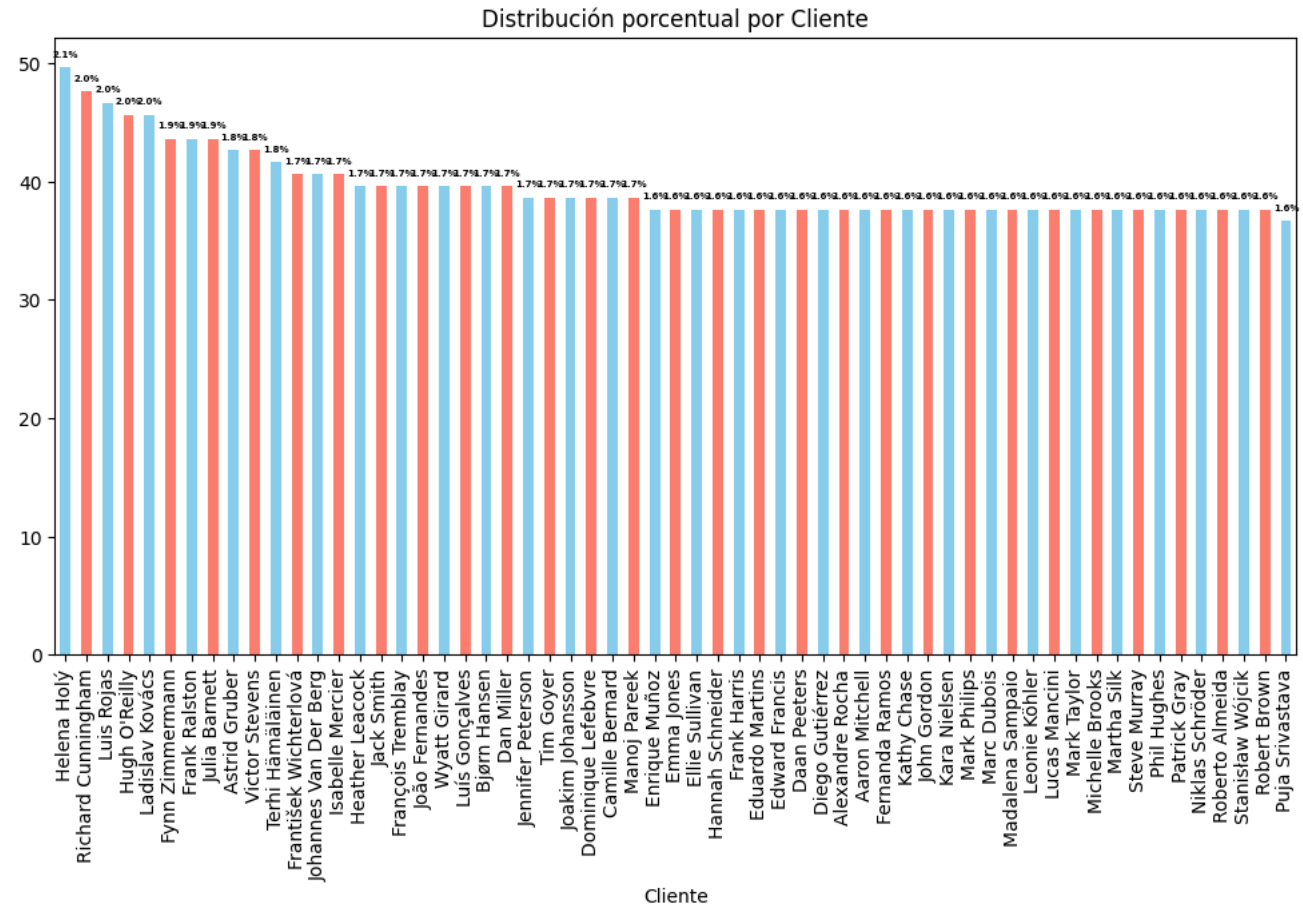
```
                fontweight='bold')

# Toques finales
plt.title("Distribución porcentual por Género")
```

```
/tmp/ipython-input-548273541.py:4: FutureWarning: The default of observed=False is deprecated and will be changed to True in a f
  data_genero = df_master.groupby("Genero")["Total_Linea"].sum().sort_values(ascending=False)
Text(0.5, 1.0, 'Distribución porcentual por Género')
```



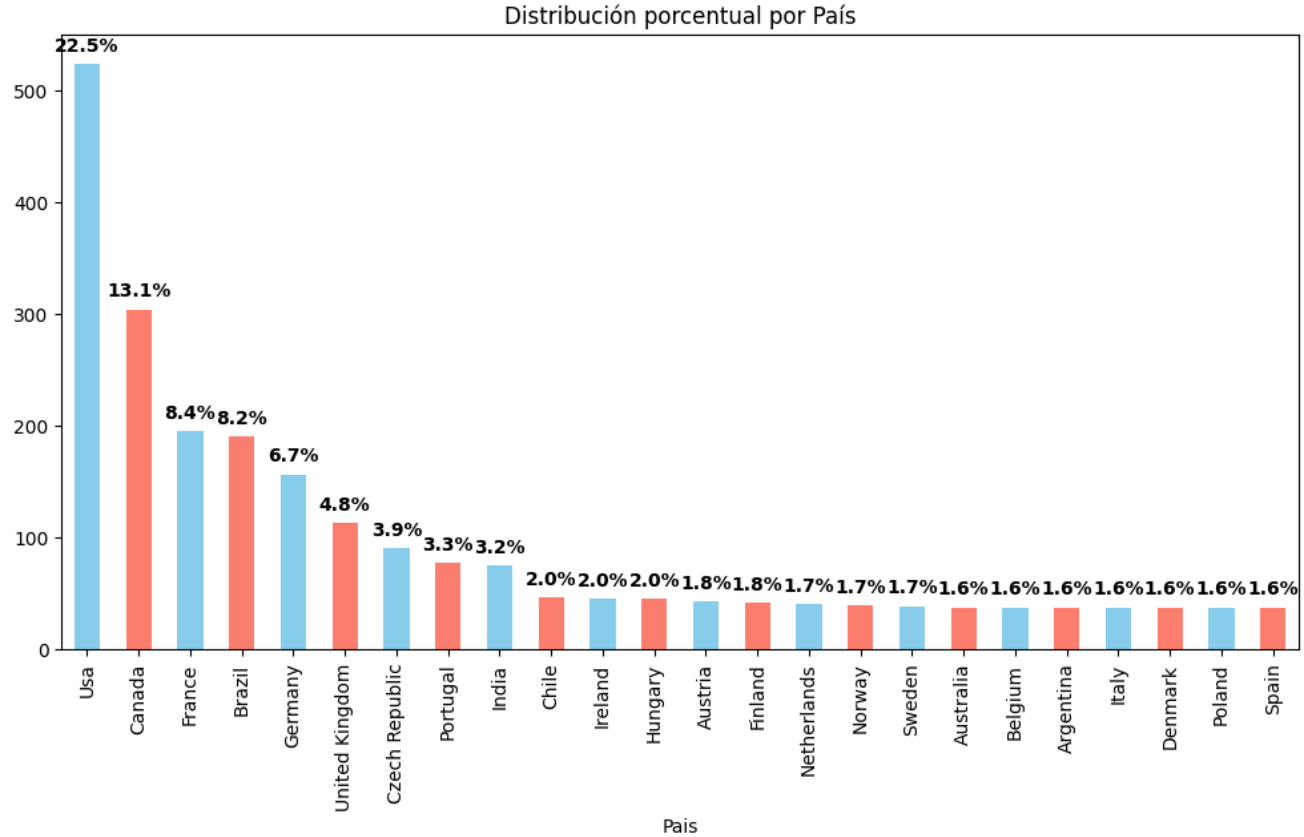

```
#graficamos
# 1. Agrupamos y calculamos el total general
data_cliente = df_master.groupby("Cliente")["Total_Linea"].sum().sort_values(ascending=False)
total_general = data_cliente.sum()

# 2. Creamos el gráfico
ax = data_cliente.plot(kind="bar", figsize=(12, 6), color=['skyblue', 'salmon'])

# 3. Iteramos para poner los porcentajes
for p in ax.patches:
    # Calculamos el % (valor de la barra / total)
    porcentaje = f'{(100 * p.get_height() / total_general):.1f}%'

    # Anotamos el texto sobre la barra
    ax.annotate(porcentaje,
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='bottom',
                xytext=(0, 5),
                textcoords='offset points',
                fontweight='semibold',
                fontsize=5)

# Toques finales
plt.title("Distribución porcentual por Cliente")
```

```
Text(0.5, 1.0, 'Distribución porcentual por Cliente')
```




```
# 1. Agrupamos y calculamos el total general
data_pais = df_master.groupby("Pais")["Total_Linea"].sum().sort_values(ascending=False)
total_general = data_pais.sum()

# 2. Creamos el gráfico
ax = data_pais.plot(kind="bar", figsize=(12, 6), color=['skyblue', 'salmon'])

# 3. Iteramos para poner los porcentajes
for p in ax.patches:
    # Calculamos el % (valor de la barra / total)
    porcentaje = f'{(100 * p.get_height() / total_general):.1f}%'

    # Anotamos el texto sobre la barra
    ax.annotate(porcentaje,
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='bottom',
                xytext=(0, 5),
                textcoords='offset points',
                fontweight='bold')

# Toques finales
plt.title("Distribución porcentual por País")
```

```
/tmp/ipython-input-2908615272.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a
  data_pais = df_master.groupby("Pais")["Total_Linea"].sum().sort_values(ascending=False)
Text(0.5, 1.0, 'Distribución porcentual por País')
```




```
df_master.groupby("Artista")["Total_Linea"].sum().sort_values(ascending=False).head(10).plot(kind="bar")
```

```
/tmp/ipython-input-913065956.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a f
```